

Milestone 1 - Initial Data Inspection & Profiling

Part of User Churn Project

OVERVIEW

The Waze data analytics team is actively executing a data-driven project to improve user retention by analyzing and predicting monthly user churn. For this initiative, churn is defined as a user uninstalling the application or completely ceasing activity within the month.

PROJECT STATUS

Milestone 1 - Initial Data Inspection & Profiling

- **Target Goal:** Audit and organize the newly accessed user dataset to evaluate its structural readiness for predictive modeling.
- **Methods:**
 - Programmatically ingested the source dataframe to evaluate shape and schema.
 - Isolated structural anomalies and quantified missing records across all features.
 - Generated foundational summary statistics to assess user baseline behaviors.

NEXT STEPS

- It would be highly useful to investigate how this data was collected to ensure we aren't looking at a non-random sample.
- Conduct EDA and Data Visualization to understand the data distribution

KEY INSIGHTS

- The source file contains 14,999 rows representing distinct users across 13 different variables.
- A total of 700 records are missing entries in the **label** column, the percentage of iPhone nulls are 63.86% and Android nulls are 36.14
- Out of 14299, the % of retained are 82.26, and the % of churned are 17.74
- Users who churned averaged ~3 more drives in the last month than retained users, but retained users used the app on over twice as many days as churned users in the same time period.
- The median churned user drove ~200 more kilometers and 2.5 more hours during the last month than the median retained user.
- The median user who churned drove 698 kilometers each day they drove last month, which is almost ~240% the per-drive-day distance of retained users. The median churned user had a similarly disproportionate number of drives per drive day compared to retained users.
- The ratio of iPhone users and Android users is consistent between the churned group and the retained group, and those ratios are both consistent with the ratio found in the overall dataset.